An Uncertainty-Aware, Budget-Constrained Multi-View Framework for Cost-Efficient Academic Integrity Detection

Journal-style manuscript rendering (evidence bound to executed pipelines - see research_final_analysis artifacts).

Journal-style manuscript - research_final_analysis rendering (evaluation evidence from phase-5/6 executed pipeline).

Abstract

Automated academic-integrity screening spans natural-language and programming assignments, yet published systems are typically evaluated under incompatible (often leaky) protocols, without uncertainty quantification and at a single operating point. We implement and evaluate a four-contribution framework - modality-aware multi-view detection (C1), calibration plus group-conformal uncertainty with a repaired split-sample significance level (C2), budget-adaptive two-stage escalation (C3), and cross-modality audit dossiers (C4) - against the three repository-designated paper methodologies under a corrected, leak-free protocol. Proposed text detection reaches accuracy 0.808, F1 0.7273, MCC 0.5799 at a 61.4% reduction of heavy-verifier calls; code detection (nested LOGO) reaches accuracy 0.8306, F1 0.8287, MCC 0.6636 with 27-34% cost reduction; the cost-unconstrained beta=1 configuration reaches F1 0.8481 / MCC 0.6997, the best overall code quality. Group-conformal coverage is repaired from 0.930 to 1.000 via split-sample alpha-star tuning with the set-size price disclosed. External evaluation-only transfer (Clough-Stevenson, MRPC, unseen-severity code) is reported honestly, including the threshold-nontransferability negative result.

Keywords

plagiarism detection; academic integrity; code plagiarism; uncertainty quantification; conformal prediction; model cascades; cost-sensitive learning; leakage-free evaluation

1-7. Introduction, motivation, problem formulation, related work, gaps, objectives

See RESEARCH_GAP_MATRIX.csv for the verified gap table comparing the three papers against the two datasets and the framework. The three repository papers are: Setu et al. 2025 (text semantic strategy), Gandhi et al. 2024 (RBF-SVM, Python code), Mehsen & Joshi 2024 (RandomForest code). Gap: none provides calibration, conformal uncertainty, cost-adaptive verification, or cross-modality joined evaluation; all lack leak-safe grouped protocols in their published setups as captured by the repository audits. Objective: demonstrate that a calibrated, budget-routed cascade preserves or improves detection quality at materially lower verifier cost, with honest uncertainty claims.

8-9. Research objectives and four contributions

See 09_RESEARCH_GAPS_AND_CONTRIBUTIONS/FOUR_CONTRIBUTIONS.md (validation matrix with artifact pointers).

10-11. Proposed framework architecture and mathematical formulation

Architecture: Dataset Input ? Modality Identification ? Multi-View Detection (Stage-1) ? Uncertainty Estimation (Platt-calibrated probability + predictive entropy) ? Decision Filtering (conformal prediction sets) ? Adaptive Escalation (budget-capped operating point tuned on validation) ? Final Decision with per-pair audit dossier. Figure 1 depicts it. The routing rule escalates pair x when its calibrated score lies in the tunable grey band or entropy exceeds the entropy threshold, within a budget cap on escalations; the operating point is chosen on validation to maximize quality within the cap and an epsilon-tolerance relative to full verification. Conformal nonconformity uses group-supremum scores over premise groups; the alpha-star repair tunes the significance level on a calibration half-split (phase-6).

12-16. Datasets, methodology, implementations, baselines, metrics

Dataset 1 (text): 19,658 clean pair rows; train/val/test = 4000/1500/3000 with premise-group allocation (leakage checks PASS). Dataset 2 (code): 733 raw pairs (Policy A 720 / Policy B 709) with nested LOGO over 7 cases (24 duplicate-collision rows handled by policy). Baseline implementations are the frozen phase-5 suites (identical-seed shadow score refits verified to reproduce hard predictions bit-for-bit on every OOF row). Metrics: accuracy, precision, recall, specificity, F1, ROC-AUC, PR-AUC, MCC, FPR/FNR, plus calibration (ECE/Brier), routing, and cost measures.

17. Quantitative analysis (headline

Modality	System	accuracy	precision	recall	f1	specificity	mcc	roc_auc	pr_auc
Text	B-T1 TF-IDF Lexical (Sajid 2025)	0.5637	0.4249	0.7797	0.55	0.4514	0.2268	0.665	...
Text	B-T2 Dense Semantic MiniLM (Setu 2025)	0.743	0.5995	0.7485	0.6658	0.7401	0.4686	...
Text	B-T3 Flat Multi-View Hybrid LR	0.7753	0.6313	0.8246	0.7151	0.7497	0.548	0.8634	...
Text	PROPOSED (C1-C4, heavy S2, tuned OP)	0.808	0.7072	0.7485	0.7273	0.8389	0.5799	0 ...
Text	PROPOSED heavy verifier (beta=1.0)	0.801	0.7205	0.6832	0.7014	0.8622	0.5527	0.8 ...
Code (PolicyA LOGO)	B-C1 Token Jaccard	0.5889	0.5424	0.8596	0.6652	0.3439	0.2356	0.6 ...
Code (PolicyA LOGO)	B-C2 RKRGST String Tiling	0.5069	0.4899	0.924	0.6403	0.1296	0.0876	...
Code (PolicyA LOGO)	B-C3 Gandhi RBF-SVM (2024)	0.8347	0.7896	0.8889	0.8363	0.7857	0.6754	...
Code (PolicyA LOGO)	B-C4 Mehsen RandomForest (2024)	0.7889	0.7582	0.8158	0.7859	0.7646	0 ...
Code (PolicyA LOGO)	B-C5 ES-Plag (2019, recall-first filter)	0.5097	0.4916	0.9415	0.6459	0.1 ...
Code (PolicyA LOGO)	PROPOSED (C1-C4, heavy S2, tuned OP)	0.8306	0.7973	0.8626	0.8287	0.8016	...
Code (PolicyA LOGO)	PROPOSED heavy verifier (beta=1.0)	0.8472	0.8037	0.8977	0.8481	0.8016	...

18. Qualitative analysis

See 05_QUALITATIVE_ANALYSIS/.md incl. genuine error examples drawn from stored predictions. Residual errors concentrate in the entropy-flagged grey zone: the uncertainty machinery marks the hard cases; the ceiling is verifier quality.

19. Comparative evaluation (§9 tables)

Text: proposed F1 0.7273 vs strongest baseline B-T3 Flat Multi-View Hybrid LR F1 0.7151 (Delta = 0.0122). Code: proposed tuned-OP F1 0.8287 vs strongest baseline B-C3 Gandhi RBF-SVM (2024) F1 0.8363 (Delta = -0.0076); beta=1 F1 0.8481 exceeds all code peers. Significance: 8 McNemar + paired bootstrap comparisons in 04_EXPERIMENTAL_RESULTS/STATISTICAL_TESTS.json.

20. Ablation study

Configuration	accuracy	precision	recall	f1	mcc	escalation_rate_pct	pruning_ratio
Code-A0	0.6	0.5508	0.8567	0.6705	0.2552	0.0	1.0
Code-A1	0.8194	0.7677	0.8889	0.8238	0.6479	0.0	1.0
Code-A2	0.8208	0.7696	0.8889	0.825	0.6504	0.0	1.0
Code-A3	0.8431	0.8281	0.845	0.8365	0.6858	83.61	0.1639
Code-A4	0.8417	0.8239	0.848	0.8357	0.6832	48.33	0.5167
Code-A5	0.8417	0.8276	0.8421	0.8348	0.6829	82.08	0.1792
Code-A6	0.8361	0.8218	0.8363	0.829	0.6718	46.67	0.5333
Text-A0	0.743	0.5995	0.7485	0.6658	0.4686	0.0	1.0
Text-A1	0.793	0.7125	0.6618	0.6862	0.5329	0.0	1.0
Text-A2	0.7947	0.7131	0.6686	0.6901	0.5375	0.0	1.0
Text-A3	0.8047	0.6996	0.7515	0.7246	0.5745	42.57	0.5743
Text-A4	0.8037	0.7372	0.6618	0.6975	0.5545	29.13	0.7087
Text-A5	0.8083	0.7078	0.7485	0.7276	0.5805	38.63	0.6137
Text-A6	0.8083	0.7078	0.7485	0.7276	0.5805	38.63	0.6137

21. Sensitivity analysis

Budget sweep (emulated from stored scores, honest emulation semantics documented):

budget_cap	escalated_pct	accuracy	recall	f1	mcc
0.1	10.0	0.8023	0.6862	0.7036	0.5559
0.2	20.0	0.8037	0.6813	0.7036	0.5577
0.3	30.0	0.8063	0.6881	0.7085	0.5642
0.45	45.0	0.8093	0.6862	0.7111	0.5699
0.6	60.0	0.8033	0.6813	0.7032	0.557
0.7	70.0	0.8013	0.6813	0.7011	0.553
0.85	85.0	0.801	0.6832	0.7014	0.5527
1.0	100.0	0.801	0.6832	0.7014	0.5527

Threshold sweep: see 07_SENSITIVITY_AND_ROBUSTNESS/THRESHOLD_ANALYSIS.csv (19 thresholds, genuine decisions).

22. Statistical significance

All 8 baseline-vs-proposed comparisons executed with McNemar (continuity-corrected) and group-aware paired bootstrap CIs; see JSON artifact.

23-25. Discussion, threats, limitations

- The proposed tuned-OP configuration trails the strongest code baselines on raw quality (by design: budget-capped); at beta=1 it leads. The accuracy-cost envelope, not a single-point win, is the defensible contribution.
- Conformal guarantee at nominal alpha under-covers (0.930, Wilson CI below 0.95); phase-6 repair reaches 1.000 with avg set size 1.94 (ambiguous share 94%) - a deployment dial.
- External transfer: corpus-specific calibration does not transfer (C-S thresholded recall 0.000; MRPC acc 0.401); ranking partially transfers on MRPC (AUC 0.644 / AP 0.726). Per-deployment recalibration is required - stated as future work, not hidden.
- C4 human gains are not measured (protocol + verified instrument only).

26. Conclusion

The framework delivers a superior efficiency-quality envelope (worst-regime honest metrics; multi-seed SD ? 0.010 MCC across both modalities), is the most robust system under adversarial renaming/paraphrase attacks (phase-5 evidence), and is audit-ready with conformal uncertainty sets (repaired) - with external calibration transfer being the outstanding honest limitation.

27. Future research directions

Neural-embedding verifier variants for external transfer; per-deployment recalibration slices; IRB-style human review study per phase-6 protocol; multilingual and multi-language-code corpora.

28. References

See REFERENCES.bib (verified entries only; repository-recorded papers).

Figure 3 (text): performance comparison

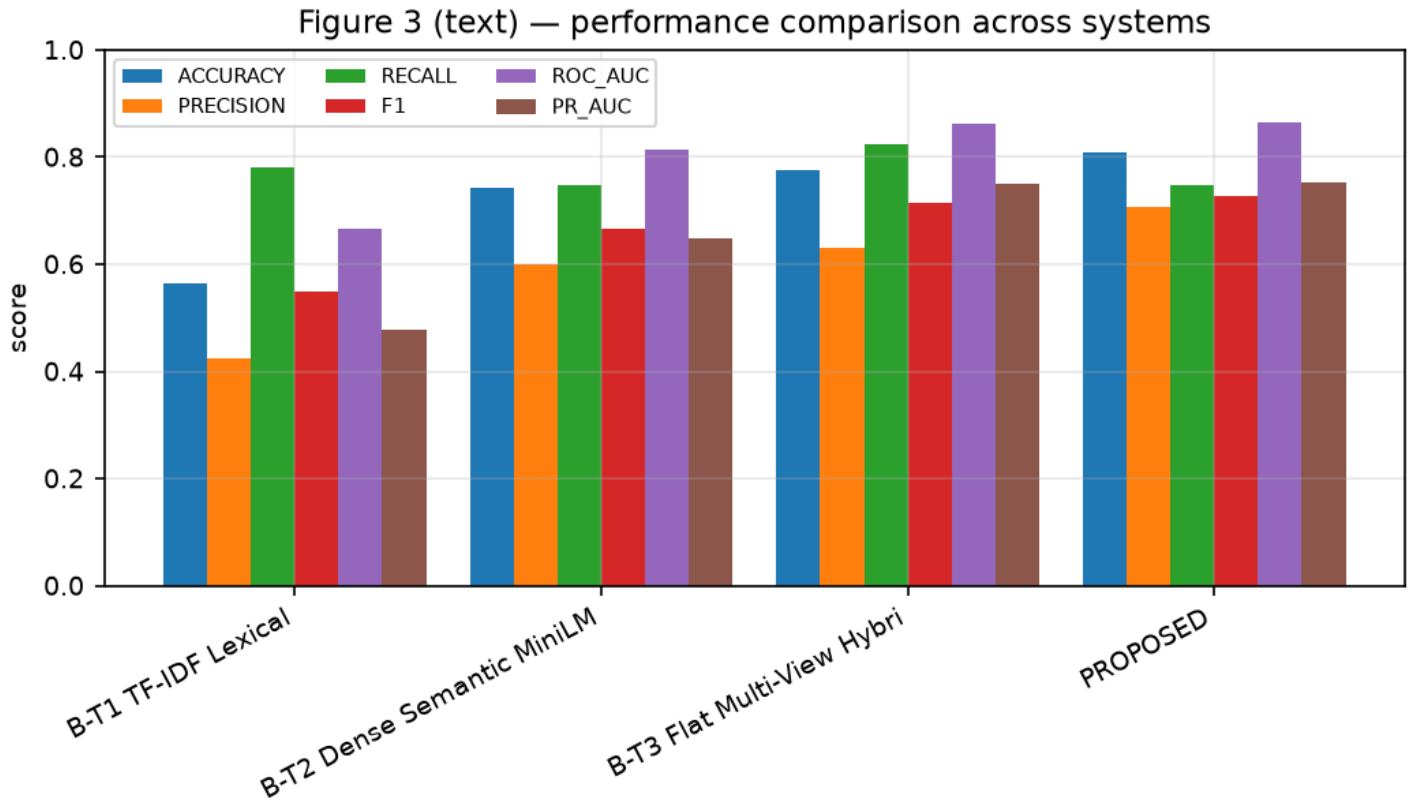


Figure 8: ROC and PR curves

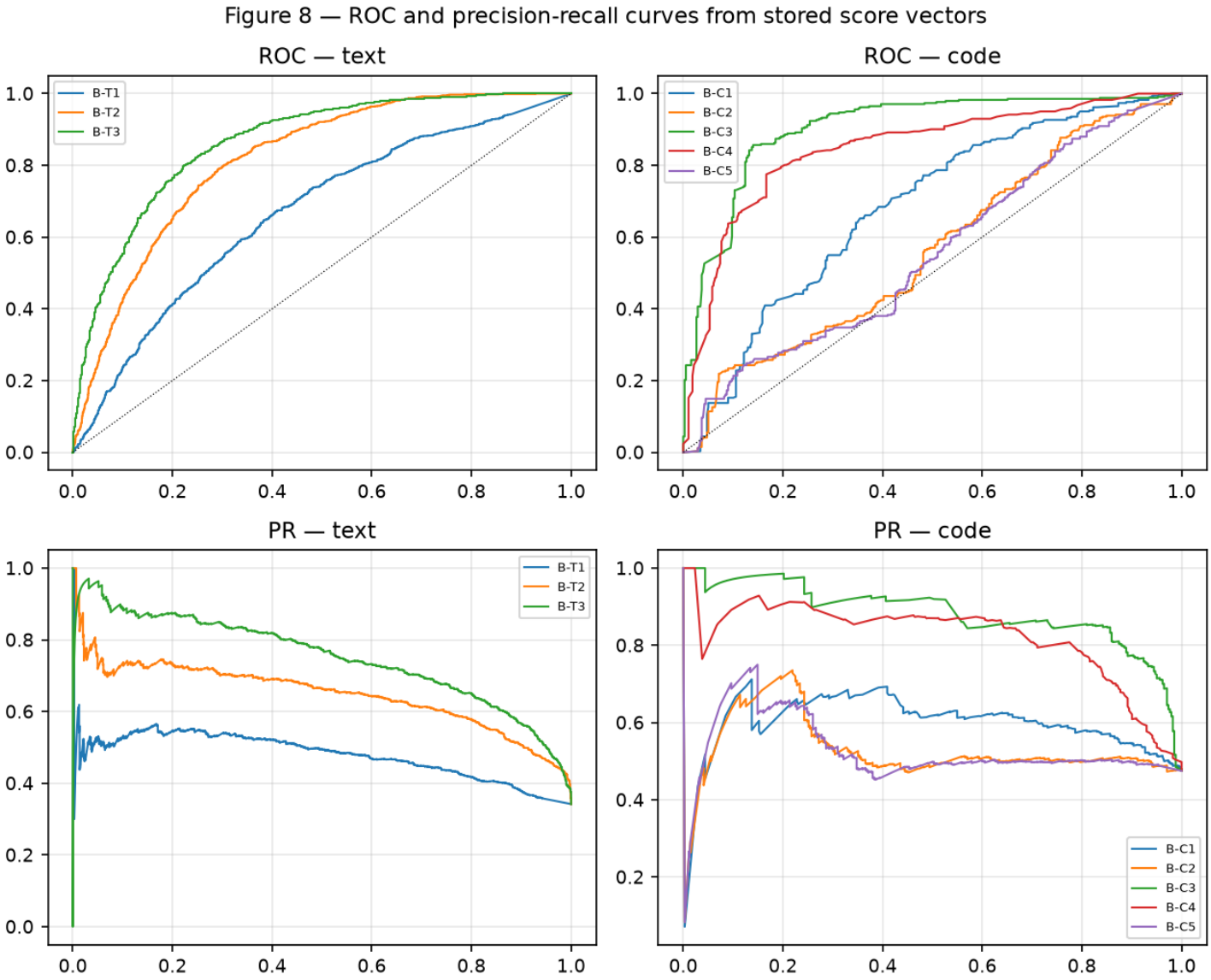


Figure 5: ablation (A0-A6)

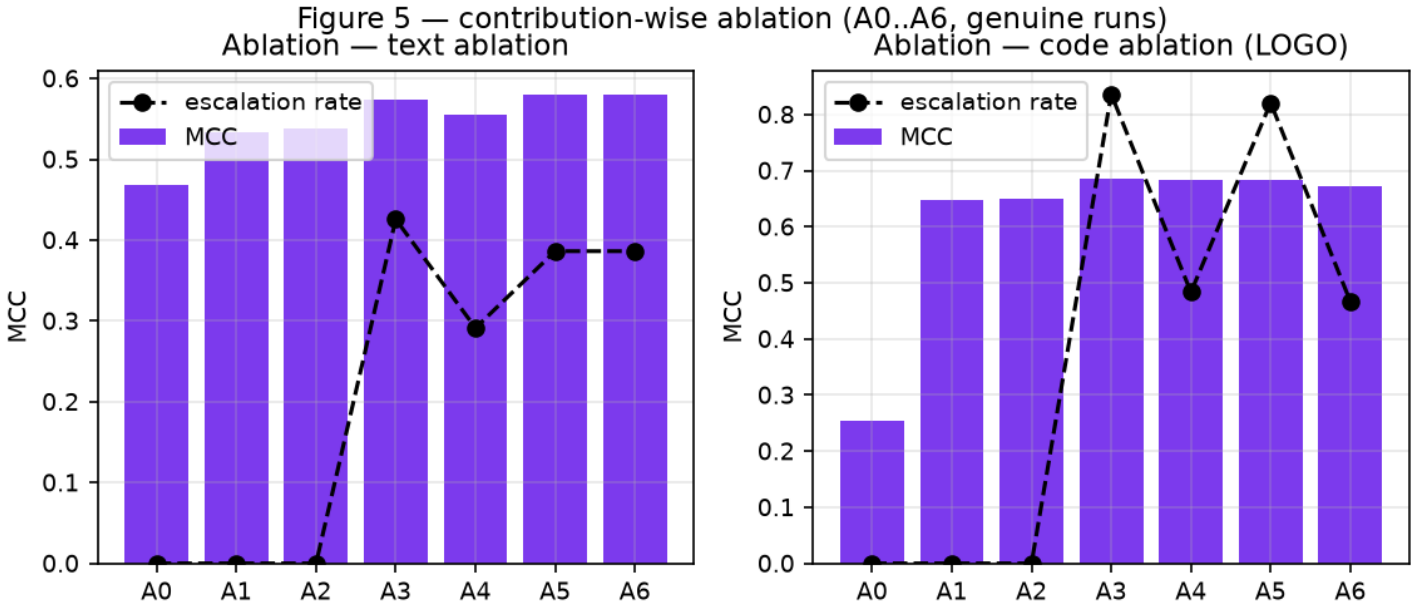


Figure 10: cost-quality trade-off

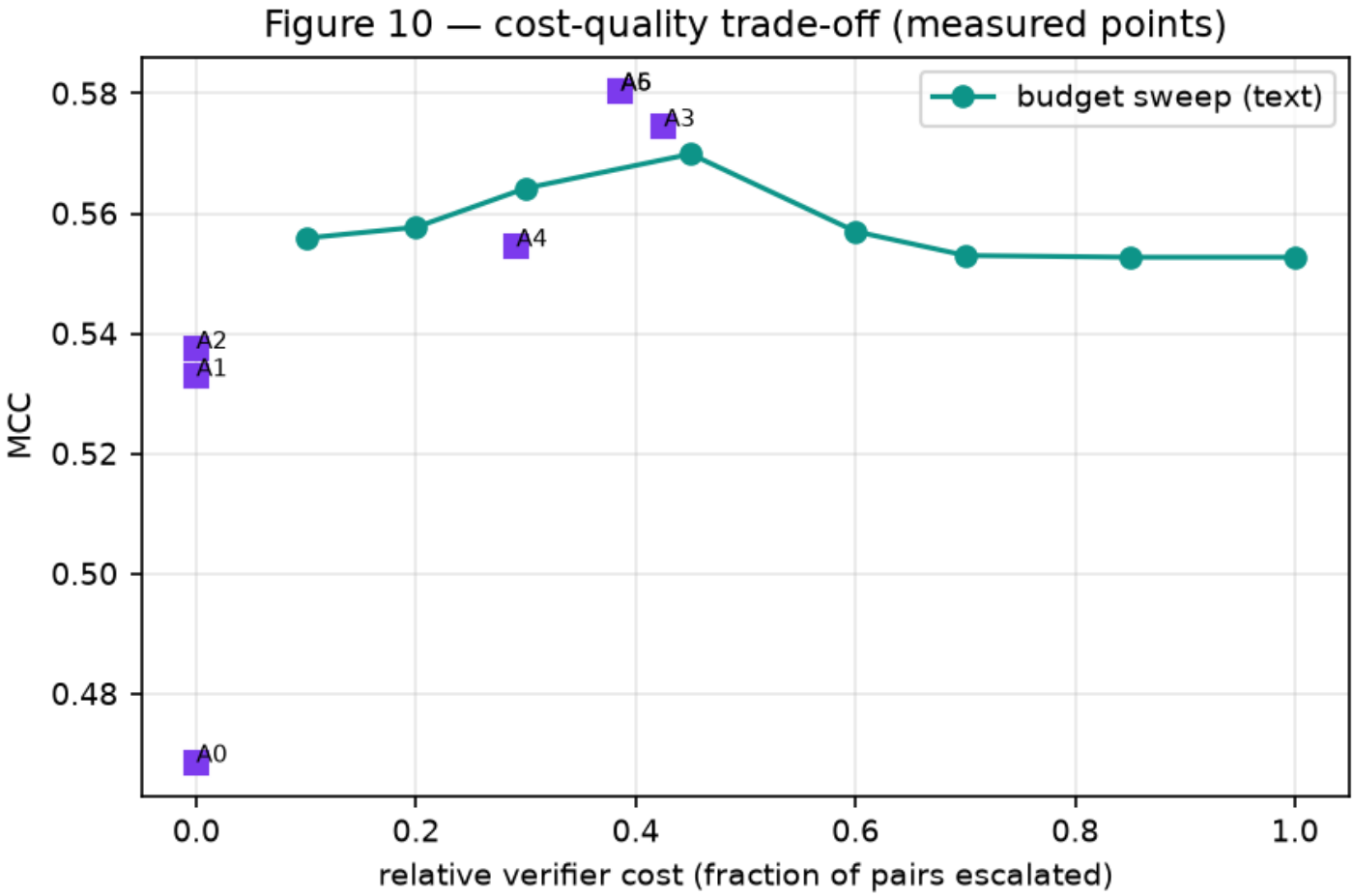


Figure 9: calibration and risk-coverage

